PWS CUP 2026 Main cohort 28 / Preliminary cohort 5

Steteco Is Embarrassing but Useful

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Rare MIA: Scoring Oracle — Score 1.0000

00 Overview: mean *TPR* → *TPR* combinations → member counts per team → individual members

Aim Identify rare records in each team's original B_i that are shared with our B_{28}.

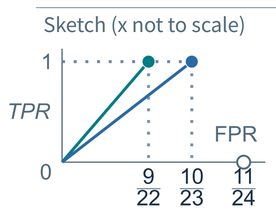
Known CodaBench rare mean *TPR* (mia.tpr_rare); 13 rare records shared by aia_challenge_28 and B_{28} are known nonmembers.

R1–2 2 submissions → member counts ➡ **R3–7** 5 oracle submissions → match → members ➡ **R8** Submit the identified members

01 Rounds 1–2: Identify member counts

Aim Find each team's rare-member count.

1.1 Tie all 11 candidates for first place...



2 members (9/22, 2/2)
TPR@1% = (1/100)×(2/2)÷(9/22) = 11/450 ≈0.02444…

1 member (10/23, 1/1)
TPR@1% = (1/100)×(1/1)÷(10/23) = 23/1000 = 0.023

0 members (excluded from rare mean *TPR*)
Point (11/24, —); *TPR*@1% = 0/0: undefined

1.2 Add known nonmembers to the top tie group

Known nonmembers in the top tie group change *TPR*.

Tie size / members	11	12	⋯	24
1	0.0230	0.0209	⋯	0.0100
2	0.0244	0.0220	⋯	0.0100

1.3 Vary tie sizes across teams

Adding 0–13 known nonmembers gives 14 possible tie sizes.
Apply these tie-size settings across 22 teams.

1.4 Infer *TPR* combinations from their mean

Match the 18-decimal mean *TPR*: 0–2 per team, ≤11 total.
22 teams share 14 settings; multiple assignments remain.
Vary settings in round 2; jointly, one assignment remains here.

1.5 Identify team counts from two submissions

Round	Assignments (teams with 1 / 2 members)	Total
1	1: {(3 or 6),11,17} / 2: {10,16,25} 1: {11,12,22} / 2: {10,13,25}; others	102
2	1: {(2 or 6 or 14 or 26),11,17} / 2: {10,16,25} 1: {3,6,11} / 2: {4,13,25}; others	52

Unique: 1→{6,11,17}; 2→{10,16,25} [6 teams, 9 members]

02 Rounds 3–7: Identify members in 5 submissions

Aim Identify each team's members in the remaining 5 submissions.

2.1 One member: 4 suffice; two: ambiguity remains

Select some of the 11 candidates for the top-confidence tie group.
Give unselected candidates zero; change the selection each time.

One member: 4 binary outcomes distinguish all 11 candidates.
$2^4 - 1 = 15 > 11$ (exclude the all-zero code)
Two members: distinct pairs can have the same count sequence.
0001 + 0110 = 0010 + 0101 = (0,1,1,1)
We need a unique member assignment,
so this scheme may not suffice in 4 submissions.

2.2 Fixed 5-round codes distinguish singles and pairs

Conditions to distinguish 11 singles and C(11,2) = 55 pairs

① **All 11 individual codes differ.**
② **All 55 elementwise pair sums differ.**

ID	3	4	5	6	7
47	0	1	1	1	1
76	0	0	0	1	1
90	0	0	1	0	1
115	1	1	0	0	1
169	1	0	0	0	1
264	1	0	1	1	0
304	0	1	0	1	0
383	1	0	0	1	0
462	1	1	1	0	0
495	0	0	1	0	0
675	0	1	0	0	0

Fixed 5-round code table (satisfies ① and ②)
1 = top tie group; 0 = zero score. Select 5 each time.

2.3 Read 6 team counts from the mean *TPR*

Add known nonmembers to the 5 selected candidates.
Adjust tie sizes without changing selected-member counts.

Team	6	10	11	16	17	25
Members (known)	1	2	1	2	1	2
Tie size (set)	**6**	**7**	**7**	**11**	**10**	**12**

3 one-member teams: 2 choices each; 3 two-member teams: 3 each.
Possible count combinations per submission: $2^3 \times 3^3 = 216$.
All 216 predicted means are distinct under this design.

Round 3 mean *TPR*: 0.017203703703703704
Matching gives counts (0,1,1,1,1,1).

Use unrounded mean *TPR* to decode all 5 submissions (rounds 3–7).

2.4 Match 5-count sequences to identify all members

Team	3	4	5	6	7	Member IDs
6	0	0	1	0	1	90
10	1	1	0	2	0	304 + 383
11	1	0	0	0	1	169
16	1	1	1	1	1	76 + 462
17	1	1	0	0	1	115
25	1	1	1	1	0	264 + 675

Blue: Team6 = ID90. Green: Team10 = IDs304 + 383.
Use the counts from 01 to match singles or pairs.

5 submissions identify 9 members across 6 teams.

03 Extension: 4-round member identification

02 fixes 5 tie groups. 03 adapts the next group to the mean *TPR*.
For this count pattern, a verified decision tree identifies every one of **≈2.5 million** assignments **within 4 submissions**.
$11!/(2!)^4 = 2{,}494{,}800 \approx 2.5$ million assignments
11 candidates: 9 assigned to 6 teams, 2 assigned to none.

Method AI implementation informed by prior work [1].
Design also uses constraint solver Z3 [2].

04 Does it work for all teams?

Fixed 5: the code table in 2.2 identifies members.
4 rounds: 2.1 works for 0/1-member teams.
Some patterns with pairs are verified adaptively.
Under uniform feasible individual assignments,
≈93.3% are identifiable within 4 rounds.
Verified with the solver in 03.

05 Countermeasures

Tie handling: remove score differences caused by tie size.
Round means: do not return tiny decimal differences.
Limit submissions: reduce repeated probing opportunities.
More shared rare members: change the 0–2/team premise.
Avoid known nonmembers:
prevent cross-task matching from revealing nonmembers.

Anonymization — Utility U 0.9220

1. Anonymization and attack-risk assessment

Variant	Utility U	AIA avg.	MIA avg.
TT: Background-conditioned	0.8387	0.1034	0.0192
seo03: Rank-mix / Wald	0.9245	0.2903	0.0673
zhang11: Stratified copula	**0.9220**	**0.1091**	**0.0673** Use

TT: Synthesize from source/background data
seo03: Mix 3 records' ranks; adjust Wald statistic
zhang11: Stratify; calibrate regular outcomes
Evaluate 15 variants with 12 AIA / 4 MIA attacks

2. Generating zhang11

Step1 Split into rare and regular strata using the official Mahalanobis distance.
Step2 Regenerate attributes via Gaussian copula.
Step3 Calibrate regular outcomes with event/death counts and Cox regression.

3. Refining zhang11

Apply one TG swap selected by hill climbing.

index	age	⋯	TG	⋯	*time*	onset	death
315	73	⋯	51.15		0.75	0	1
1048	73	⋯	156.02		5.4	0	1

TG swap comparison	Before	After
Mean AIA	**0.1531**	**0.1397**
Utility U	0.9220	0.9220

Mean AIA before → after TG swap
0.1531 → 0.1397 (about 8.8% lower)

Overall MIA — Score 0.0715

Input: our original B_{28}, public A_{bg} and anonymized C_i.
Choose a method per target and score each candidate.
Member: also in private B_i. A_{bg} records are nonmembers.

1. Gen-LRA: Add a candidate and compare [3]

Before adding	After adding
Distribution fitted to A_{bg}	Fit to A_{bg} + 1 candidate

Score the same 10 nearby C_i rows in both distributions.
Sum log-density differences (a measure of likelihood).

A larger total difference gives a higher candidate score.

Nonmember calibration: use public AIA nonmembers;
correct score bias by prefecture and sex.

2. Nearest-neighbor ratio using 12 attributes

Find each dataset's nearest row using 12 attributes.

Public data	Candidate	Anonymized
A_{bg}	(B_{28})	C_i

Nearest distance to A_{bg} — Nearest distance to C_i

Use the distance ratio to score membership likelihood.
Relatively closer to C_i → higher candidate score.

3. Choose methods per attack target

Use method 2 only for targets where it appears effective.
Exclude the record itself to avoid zero self-match distance.

Nonmember baseline	Compare with baseline
A_{bg} rows → method 2, excluding self-matches.	Count B_{28} candidates well above the baseline.

Multiple candidates	Other targets
2. Nearest-neighbor ratio	1. Gen-LRA + calibration

Overall AIA — Score 0.2559

1. Distinguish targets from controls

$$R = \text{clip}\left(\frac{p_m - p_c}{1 - p_c}, 0, 1\right)$$

Select targets by proximity to C and background data.
Hit targets; **suppress hits on controls**.

2. Compare matched groups of 2–4

Group by prefecture, sex, assumed 5-year age band.
Example: men 60–64, same prefecture; pairs unverified.
57 groups / 181 people across 22 targets.

Pair example	Round 5	Round 6
A	**9.5 y**	11 y
B	11 y	**9.5 y**

R	Selection and comparison
5	Use preliminary target/control labels. Top ≈half: 9.5 y; rest: 11 y (per group).
6	Select by attribute-specific proximity. Swap 9.5 y / 11 y in each group.
7	Match 4 clinical attributes: SBP/TG/HDL/ALT. Per team/group, keep better round 5/6 scores.

9.5 y: A hit if time is 9–10 years.
11 y: Always wrong; avoid excluding true targets.

Round 6 also uses 0 years outside these groups.

Team	Best 1–4	Best 5–7	Round
14	0.1491	**0.4631**	6
11	0.1668	**0.4610**	6
2	0.0832	**0.3745**	5
16	0.0318	**0.2372**	7
26	0.0000	**0.2251**	7

Per-target φ-corrected AIA scores; not time accuracy.

[1] H. Verhaeghe et al. (2020). Learning Optimal Decision Trees using Constraint Programming (Extended Abstract). IJCAI, 4765–4769.
[2] L. de Moura & N. Bjørner (2008). Z3: An Efficient SMT Solver. TACAS, 337–340.
[3] Ward, Wang & Cheng (2026), arXiv:2508.21146v2. Privacy Auditing Synthetic Data Release through Local Likelihood Attacks, §4.

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